

QuoNic Algorithm Template Report

Quantum Unified Operation Native Interface Core

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77 algorithms across 10 domains. 33 full implementations + 33 minimal demos + 6 existing + 5 newly added.

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1. Overview

This report documents the expansion of QuoNic algorithm templates from 6 to 77, covering 10 domains of quantum computing.

Category	Full	Minimal	Total
Foundational	9	0	9
Search & Optimization	6	0	6
Quantum Chemistry	7	1	8
Linear Algebra	0	6	6
Communication	5	0	5
Hybrid	5	1	6
Error Correction	0	9	9
Statistical	0	3	3
Algebraic	0	3	3
Cutting-Edge	0	10	10
Existing	6	0	6
TOTAL	38	33	77

2. Foundational Algorithms

Algorithm	File	Qubits	Type	Description
QFT	qft_algo.py	n	Full	Quantum Fourier Transform
Deutsch-Jozsa	deutsch_jozsa.py	$n+1$	Full	Constant vs balanced
Bernstein-Vazirani	bernstein_vazirani.py	$n+1$	Full	Hidden bitstring
Simon	simon.py	$2n$	Full	Period finding
SWAP Test	swap_test.py	$2n+1$	Full	State overlap
Hadamard Test	hadamard_test.py	$n+1$	Full	Expectation value
Amplitude Amplification	amplitude_amplification.py	n	Full	Grover generalization
Amplitude Estimation	amplitude_estimation.py	$n+k$	Full	Grover+QPE
QPE	qpe.py	$n+k$	Full	Phase estimation

3. Search & Optimization

Algorithm	File	Qubits	Type	Description
Grover	grover.py	n	Full	Unstructured search
Quantum Counting	quantum_counting.py	n+k	Full	Count solutions
QAOA Generic	qaoa_generic.py	n	Full	User-defined Hamiltonian
QAOA MaxCut	qaoa.py	n	Full	MaxCut
QAOA TSP	qaoa_tsp.py	n^2	Full	Traveling salesman
QAOA MIS	qaoa_mis.py	n	Full	Maximum independent set
QAOA Knapsack	qaoa_knapsack.py	n	Full	Resource allocation
Quantum Walk	quantum_walk.py	n+1	Full	Cycle graph walk
Quantum Annealing	quantum_annealing.py	n	Minimal	Simulated annealing

4. Quantum Chemistry

Algorithm	File	Qubits	Type	Description
VQE	vqe.py	n	Full	Variational eigensolver
VQE Molecular	molecule_vqe.py	n	Minimal	H2 molecule
Hamiltonian Import	hamiltonians_ext.py	-	Full	OpenFermion/PennyLane
Trotter	trotter.py	n	Full	Time evolution
Hamiltonian Sim	hamiltonian_simulation.py	n	Full	Heisenberg model
Dynamics Sim	dynamics_simulation.py	n	Full	Time-dependent $H(t)$
Fermion Mapping	fermion_mapping.py	-	Full	Jordan-Wigner
QSP	qsp.py	1	Minimal	Signal processing

5. Linear Algebra (all minimal)

Algorithm	File	Qubits	Type	Description
HHL	hhl.py	$2+k+1$	Minimal	2x2 diagonal matrix
Matrix Inversion	matrix_inversion.py	$2+k+1$	Minimal	HHL special case
Eigenvalue Solver	eigenvalue_solver.py	$n+k$	Minimal	QPE application
PDE Solver	quantum_pde.py	n	Minimal	1D heat equation
ODE Solver	quantum_ode.py	n	Minimal	Exponential decay
Data Fitting	quantum_fitting.py	1	Minimal	Quantum least squares

6. Communication & Cryptography

Algorithm	File	Qubits	Type	Description
Teleportation	teleportation.py	3	Full	State transfer
BB84	bb84.py	1	Full	QKD protocol
E91	e91.py	2	Full	Entanglement QKD
Superdense Coding	superdense_coding.py	2	Full	2 bits via 1 qubit
Shor	shor.py	2n	Full	Integer factoring
Discrete Log	discrete_log.py	-	Minimal	Small modulus

7. Hybrid Quantum-Classical

Algorithm	File	Qubits	Type	Description
VQC	vqc.py	n	Full	Variational classifier
Quantum Kernel	quantum_kernel.py	$2n+1$	Full	SWAP test kernel
QNG	qng.py	n	Full	Natural gradient
VQR	vqr.py	1	Full	Variational regression
QNN	qnn.py	n	Minimal	Generic variational circuit
QSVM	qsvm.py	$2n+1$	Minimal	Quantum kernel + SVM
QA Hybrid	quantum_annealing_hybrid.py	-	Minimal	Classical + quantum

8. Error Correction (all minimal)

Algorithm	File	Qubits	Type	Description
Bit-flip Code	bit_flip_code.py	3+2	Minimal	3-bit repetition
Phase-flip Code	phase_flip_code.py	3+2	Minimal	H + bit-flip
Shor 9-bit	shor_code.py	9+4	Minimal	Full error correction
Steane 7-bit	steane_code.py	7+6	Minimal	CSS code
Stabilizer	stabilizer.py	n	Minimal	Clifford group
Syndrome	syndrome.py	n+k	Minimal	Error detection
Surface Code	surface_code.py	5+4	Minimal	3x3 lattice
Color Code	color_code.py	7	Minimal	7-bit color code
FT Gates	ft_gates.py	2	Minimal	T gate magic state

9. Statistical & Sampling (all minimal)

Algorithm	File	Qubits	Type	Description
Quantum Monte Carlo	quantum_monte_carlo.py	n	Minimal	Amplitude estimation
Rejection Sampling	rejection_sampling.py	n	Minimal	Grover-based
Quantum Bayesian	quantum_bayesian.py	1	Minimal	Hypothesis testing

10. Algebraic & Hidden Subgroup (all minimal)

Algorithm	File	Qubits	Type	Description
HSP	hsp.py	$2n$	Minimal	Abel HSP (Simon)
Lattice SVP	lattice.py	-	Minimal	2D shortest vector
Elliptic Curve	elliptic_curve.py	-	Minimal	Small curve DLP

11. Cutting-Edge Demos (all minimal)

Algorithm	File	Qubits	Type	Description
QCNN	qcnm.py	2	Minimal	Quantum CNN
QGNN	qgmn.py	3	Minimal	Quantum GNN
DQAOA	dqaoa.py	4	Minimal	Distributed QAOA
QTransformer	qtransformer.py	1	Minimal	Self-attention
QRL	qrl.py	1	Minimal	Reinforcement learning
QTDA	qtda.py	1	Minimal	Topological analysis
QPCA	qpca.py	1	Minimal	PCA
Clustering	quantum_clustering.py	-	Minimal	k-means
QGAN	qgan.py	1	Minimal	GAN
QBM	qbm.py	2	Minimal	Boltzmann machine

12. Boundary Conditions

Area	Constraint
General	All on simulators; ~20 qubit limit; statistical noise; 4^n DM memory
VQE/QAOA	StatevectorSimulator for exact expectations, not backend sampling
SWAP Test	Native cswap; CCX decomposition broken for superposition targets
Error Correction	Minimal demos; full FT needs 100+ qubits
HHL/PDE/ODE	Minimal demos; production needs complex implementation
Cutting-edge	Concept demos, not production-grade
Cqlib	No local simulator; cloud-only (TianYan)
MindQuantum	No Windows wheel; Linux/macOS only
CUDA-Q	CUDA 13.1 too new; needs specific CUDA version
Paddle Quantum	Removed: dependency conflicts on Python 3.13+

13. Technical Decisions

Decision	Rationale
CSWAP gate	Added native cswap to IR; CCX decomposition broken for superposition
Bit ordering	qubit 0 = rightmost (LSB); auto-measure all unmeasured qubits
Full vs Minimal	Full = standard algorithm, works on simulator. Minimal = concept only
Paddle Quantum	Removed: Paddle 3.x dependency conflicts on Python 3.13+
Backend strategy	EngineBackend preserved; no Translator for engine backends
Global state	No changes; parallelism via ProcessPoolExecutor

Test results: 379 passed, 38 skipped, 0 failed